HOSSEIN AKHLAGHPOUR

Principal Applied AI Engineer | Staff AI Engineer | Enterprise Context & LLM Systems Architect

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EXECUTIVE SUMMARY

Principal-level applied AI and machine learning engineer with 8+ years building production AI/ML systems and 10+ additional years in software architecture, distributed systems, and technical leadership. Deep hands-on experience designing and shipping enterprise NLP, GenAI, natural-language interfaces, information extraction, secure API/tool orchestration, ML pipelines, and AI-ready context systems for legal, CRM, and private-capital / financial-services workflows. Strong fit for Staff / Principal AI roles requiring both architecture ownership and implementation depth across LLM applications, RAG, context engineering, schema grounding, evaluation, guardrails, production reliability, cloud infrastructure, and cross-functional delivery.

TARGET ROLES

Principal AI Engineer · Staff AI Engineer · Principal Machine Learning Engineer · Staff Machine Learning Engineer · Applied AI Engineer · LLM Platform Engineer · ML Platform Architect · Senior Applied Scientist / Engineer · AI Technical Lead · Enterprise AI Engineer

PRINCIPAL / STAFF-LEVEL STRENGTHS

- Designed and delivered production AI capabilities for enterprise legal, CRM, and private-capital / financial workflows, including context construction, natural-language interfaces, information extraction, compliant narrative generation, and secure API orchestration.
- Built systems that transform client data, documents, customized schemas, and workflow artifacts into AI-digestible context used for grounded legal and financial workflow automation.
- Built a natural-language CRM interface for a product used by 10,000+ customers, translating conversational questions into secure API calls and structured application outputs.
- Defined reusable engineering patterns for LLM integration, retrieval, schema grounding, context validation, tool invocation, prompt/output controls, model evaluation, guardrails, and production governance.
- Built and optimized ML systems in reliability-sensitive environments, including smart-grid systems controlling 10,000+ batteries and real-time event-dispatch pipelines.
- Designed and implemented scalable ML/data platforms using Python, Kubernetes, Spark, Kafka, AWS, REST/gRPC services, PostgreSQL, Redis, vector databases, and CI/CD workflows.
- Led teams of up to 10 developers and guided architecture decisions across ML systems, distributed infrastructure, embedded systems, cloud platforms, and high-volume data processing.

SELECTED APPLIED AI / ML SYSTEMS

Enterprise Context Engineering for Legal / Financial AI — Production Architecture

Designed and implemented approaches for transforming enterprise client data into AI-ready context for legal, CRM, and private-capital / financial-services workflows. Work included parsing structured and unstructured data, mapping customized schemas to domain concepts, preserving source traceability, grounding LLM outputs in available context, and integrating generated answers with secure product workflows.

Enterprise Natural-Language CRM Interface — Production LLM/NLP System

Architected and delivered a natural-language interface for customized CRM schemas, translating user questions into secure API calls and returning structured UI outputs plus natural-language responses. The system required schema grounding, intent interpretation, API/action generation, validation of generated operations, and safe integration into enterprise workflows.

Secure LLM Tool-Orchestration Patterns — Production Architecture

Standardized implementation patterns for context construction, model integration, prompt and output controls, tool invocation, workflow-level validation, error handling, and governance of enterprise AI features. Focused on production reliability, controlled outputs, permission-aware execution, and safe integration with existing product surfaces.

Legal Narrative Generation Workflow — Applied GenAI

Developed compliant narrative-generation workflows using historical notes, user actions, domain-specific constraints, and validation controls to automate legal documentation while preserving traceability, reviewability, and workflow safety.

Enterprise Information Extraction — Production NLP

Built an email signature extraction system using named entity recognition to convert unstructured communications into structured CRM data, improving data quality and reducing manual entry in customer/contact workflows.

Smart-Grid ML Optimization — Production ML Systems
Improved ML and optimization systems used to control more than 10,000 batteries in a smart-grid environment. Developed scalable pipelines on Kubernetes and Spark, improved event-dispatch responsiveness, and enhanced outage-detection models using gradient boosting approaches.

Agentic Enterprise Workflow Platform — Reference Architecture / Prototype
Designed a reference architecture for AI agents operating over enterprise data, APIs, documents, and human approval workflows. Covered context retrieval, tool execution, source-backed answers, workflow state, audit trails, evaluation, guardrails, and human-in-the-loop execution.

PROFESSIONAL EXPERIENCE

Intapp — Senior Machine Learning Engineer / Principal Applied AI Engineer Palo Alto, CA | Jan 2021 – Jun 2026

- Led architecture and implementation of AI-powered NLP and generative AI capabilities for enterprise legal, CRM, and private-capital / financial-services workflows.
- Helped convert client data, documents, customized schemas, and workflow artifacts into structured, AI-digestible context for LLM/NLP systems.
- Architected a natural-language interface for a CRM application used by more than 10,000 customers, enabling users to query customized schemas through natural language.
- Engineered secure translation of user intent into API calls, returning structured UI outputs and natural-language responses while respecting enterprise data models, permissions, and workflow constraints.
- Developed compliant narrative-generation workflows using historical notes, user actions, validation controls, and domain-specific output constraints.
- Built an email signature extraction system using named entity recognition to convert unstructured communications into structured CRM data.
- Standardized production engineering patterns for context construction, model integration, prompt/output controls, workflow-level validation, governance, and reliability of enterprise AI features.
- Partnered with product, engineering, and enterprise stakeholders in a forward-deployed model to translate ambiguous legal, CRM, and financial workflows into production AI systems.

AutoGrid — Senior Machine Learning Engineer Redwood City, CA | Nov 2019 – Jan 2021

- Improved ML systems used to control more than 10,000 batteries in a smart-grid environment.
- Developed scalable ML and optimization pipelines on Kubernetes and Spark to reduce AWS cost while maintaining response-time requirements.
- Enhanced a real-time event dispatcher using greedy heuristic algorithms to improve resolution speed and responsiveness.
- Improved power-outage detection using gradient boosting approaches including XGBoost and CatBoost.
- Developed a vegetation-planning and risk-identification solution using satellite imagery and U-Net segmentation models.

Vitachain — Director of Machine Learning Belmont, CA | Nov 2017 – Nov 2019

- Led machine-learning development for digital health applications focused on blood-pressure prediction and behavioral recommendations.
- Developed a contextual bandit system using Vowpal Wabbit to recommend optimal user activities based on PPG, Google Fit, and Apple Health data.
- Built a seq2seq LSTM-based model to predict blood pressure from smartwatch PPG signals.
- Partnered across product and engineering to translate health data signals into applied ML workflows and recommendation features.

Genapsys — Software Lead Redwood City, CA | Apr 2013 – Aug 2016

- Developed DNA-sequencing denoising and base-calling algorithms using PCA, Markov models, Naive Bayes, and AdaBoost.
- Led software development across embedded systems, backend services, and cloud-based distributed infrastructure supporting sequencing workflows.
- Built end-to-end systems using Docker, REST services, AWS, Kubernetes, Spark, HBase, Kafka, RabbitMQ, C++, Java, and Spring-based components.

Velti — Engineering Manager San Francisco, CA | Nov 2010 – Apr 2013

- Led a team of 10 software developers and delivered a large-scale ad analytics and conversion tracking platform.
- Supported high-volume data processing architecture handling approximately 50 billion records per month across HBase, Solr, Amazon RDS, and DynamoDB.

eBay — Senior Software Architect Brisbane, CA | Jun 2009 – Nov 2010

- Led development of a merchant reporting platform providing daily reports to more than 6,000 merchants.
- Built reporting and scheduling capabilities on top of BIRT and Quartz while maintaining system performance at scale.

Danisco — Senior Software Engineer Palo Alto, CA | Jul 2008 – Jun 2009

- Led integration with Google Enterprise Search Appliance through Lucene-based interfaces.
- Supported migration of more than 50,000 documents from IBM content management systems to Alfresco CMS.

TECHNICAL SKILLS

Applied AI / LLM Systems: LLM Applications, RAG, Context Engineering, Schema Grounding, Semantic Data Modeling, Agentic Workflows, Tool Use / Function Calling, Model Routing, Prompt Engineering, Evaluation Frameworks, Guardrails, Human-in-the-Loop Workflows, Source-Backed Answers, AI Governance, Workflow Automation

Legal / Financial AI: Legal Workflows, CRM Workflows, Private-Capital / Financial-Services Workflows, Client Data Parsing, Document Understanding, Complex Question Answering, Permission-Aware Retrieval, AI-Ready Context Construction

Machine Learning / NLP: PyTorch, Transformers, Fine-Tuning, LoRA / QLoRA, Embeddings, Reranking, NER, Information Extraction, XGBoost, CatBoost, U-Net, LSTM, Contextual Bandits, Optimization

Engineering / Platforms: Python, FastAPI, gRPC, REST APIs, Docker, Kubernetes, AWS, Spark, Kafka, Airflow, PostgreSQL, Redis, Vector Databases, CI/CD, MLflow, Kubeflow

Architecture / Leadership: System Design, ML Platform Architecture, Distributed Systems, Technical Strategy, Production Reliability, Stakeholder Alignment, Team Leadership, Cross-Functional Delivery

EDUCATION

University of Cincinnati — Master of Science in Physics